

Lab 16: Introduction to Network Traffic Analysis using Wireshark

Assignment Date: August 17th, 2026

Deadline: August 26th, 2026

Class: BSc.CSIT 4th Semester, New Summit College

Subject: Computer Network

Objective: This lab aims to introduce students to Wireshark, a powerful network packet analyzer.

Tasks:

1. Install Wireshark
2. Understand the basic features and interface of Wireshark.
3. Capture live network traffic.
4. Apply basic filters for protocol-specific packet capture.
5. Learn how to interpret the captured data.

Report Format & Structure (Topics to include for submission):

- Theory
 - a. Wireshark
 - b. Key Concepts of Wireshark
- Interface of Wireshark
 - a. Main Toolbar
 - b. Packet List Pane
 - c. Packet Details Pane
 - d. Packet Bytes Pane
- Basic Network Capture and Analysis
 - a. Selecting a Network Interface
 - b. Starting Packet Capture (For reference, use newsommit.edu.np)
 - c. Stopping and Saving Captures
 - d. Export the Captured Data
- Conclusion

References:

1. https://drive.google.com/file/d/1QVsyqy1_2p9OXv7E5bSYXGt8qH4KrZOC/view?usp=sharing